

Myo-inositol supplementation in obese patients with non-alcoholic fatty liver disease: Assessment of sirtuin-1 pathway, atherogenic and hematological parameters

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ABSTRACT

Background: Non-alcoholic fatty liver disease (NAFLD), the most common chronic liver disease, is closely associated with higher risks of atherosclerosis through a number of molecular cascades such as the sirtuin-1 (SIRT-1) pathway. Myo-inositol as a precursor isomer of cyclohexanehexols exerts anti-hyperlipidemic and anti-obesity features in metabolic disorders. The current study aimed to assess the effects of myo-inositol supplementation on genes expression of the SIRT-1 pathway, atherogenic indices, and hematological parameters in obese patients with NAFLD.

Methods: This double-blinded randomized clinical trial was conducted on 44 obese patients with ultrasound proven- NAFLD who received dietary recommendations and were randomly allocated into either myo-inositol (4 g/day) or placebo (maltodextrin 4 g/day) receiving groups for 8 weeks. At baseline and the end of the study mRNA expression levels of SIRT-1 pathway genes in peripheral blood mononuclear cells (PBMCs), complete blood cell count, lipid profile, and atherogenic indices were assessed. Inter-group differences were determined using analysis of covariance (by adjusting for the potential confounders) at the end of the trial.

Results: At the end of the trial mRNA expression levels of low-density- lipoprotein receptor (LDLR) (only in myo-inositol group) and endothelial nitric oxide synthase (eNOS) (in both groups) significantly increased, compared with steady-state levels ($p = 0.004$ and $p = 0.001$, respectively). Moreover, myo-inositol supplementation significantly inhibited lectin-like oxidized- low-density lipoprotein receptor-1 (LOX-1) expression compared with the placebo group ($p = 0.045$). Slight improvements in Cholesterol-related atherogenic indices and hematological parameters were also observed in myo-inositol group, compared with baseline values ($p < 0.05$).

Conclusion: Myo-inositol supplementation could slightly activate SIRT-1 pathway and thereby exert marginal anti-atherosclerotic properties in obese patients with NAFLD.

Abbreviations: ANCOVA, Analysis of covariance; DCI, D-chiro-inositol; eNOS, Endothelial nitric oxide synthase; FC, Fold change; HCT, Hematocrit; HDL-C, High-density lipoprotein cholesterol; Hgb, Hemoglobin; INS, Inositol; LDL-c, Low-density lipoprotein cholesterol; LDLR, Low-density- lipoprotein receptor; LOX-1, Lectin like oxidized- low-density lipoprotein receptor-1; LXR, Liver X receptor; MCV, Mean corpuscular volume; MHR, Monocyte-to-HDL-c ratio; NAFLD, Non-alcoholic fatty liver disease; OS, Oxidative stress; PBMCs, Peripheral blood mononuclear cells; PCSK9, Proprotein convertase subtilisin/kexin type 9; RBC, Red blood cell count; RDW-CV, Red cell distribution width-coefficient of variation; RT-qPCR, Reverse transcription quantitative real- time polymerase chain reaction; SD, standard deviation; SIRT-1, Sirtuin-1; TC, Total cholesterol; TG, Triglycerides; WBC, White blood cell count; NO, Nitric oxide; Non-HDL-c, non-high-density lipoprotein cholesterol; ox-LDL, Oxidized- low-density lipoprotein; ROS, Reactive oxygen species.

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1. Introduction

To date, the syndemic of chronic metabolic diseases emerges as the pioneer cause of morbidity and mortality worldwide (Chong et al., 2023). Non-alcoholic fatty liver disease (NAFLD) is defined by excessive hepatic fat deposition which is not attributed to heavy alcohol consumption (Zhang et al., 2023). As the hepatic phenomenon of multi-metabolic disorders, NAFLD affects almost 33 % of the general population and is 70 %–75 % prevalent in subjects with overweight and obesity (Riazi et al., 2022; Quek et al., 2023). NAFLD is also co-incident with hyperlipidemia, metabolic syndrome, and cardiovascular disease (Zhang et al., 2023; Arefhosseini et al., 2022). Hereditary factors, over-nutrition, insulin resistance, inflammation, and oxidative stress (OS) are the main pathological drivers of NAFLD (Powell, Wong, & Rinella, 2021). The aforementioned hits lead to an increased influx of free fatty acid to the liver and oxidized- low-density lipoprotein uptake beside mitochondrial dysfunction and over-production of reactive oxygen species which consequently trigger the progression of atherogenesis (Stols-Gonçalves, Hovingh, Nieuwdorp, & Holleboom, 2019). The formation of plaques and atherosclerotic lesions in the arterial wall are the main properties of atherogenesis, progressed by several molecular pathways related to dyslipidemia and OS (Zhang et al., 2023; Farahi et al., 2021).

Sirtuin-1 (SIRT-1), a nicotinamide adenine dinucleotide-dependent deacetylase, is involved in the deacetylation of histone/non-histone proteins and improves lipid homeostasis through its downstream genes (Feng, Liu, Xu, & Si, 2018). The SIRT-1 pathway not only exerts athero-protective, anti-oxidative, and cardio-protective functions but also is considered a metabolic sensor (Feng et al., 2018). The mentioned pathway regulates hepatic lipid and carbohydrate metabolism and thereby could be considered a therapeutic target in the management of NAFLD (Farahi et al., 2021; Feng et al., 2018).

Myo-inositol as a cyclohexanehexol and the precursor of inositol (INS) compounds, exerts a plethora of therapeutic features in metabolic disorders (Pani et al., 2020; Kiani et al., 2021; Wojciechowska et al., 2019). Myo-inositol is endogenously synthesized in eukaryotic cells and presented in foods including legumes, beans, fruits, and nuts as well as dietary supplements (Pani et al., 2020). Myo-inositol-containing molecules (such as myo-inositol phospholipids) participate in lipoproteins and cell membranes which regulate a variety of cellular functions (Wojciechowska et al., 2019). To date, myo-inositol is mostly known as an anti-diabetic agent by exerting “insulin-mimetic” properties in insulin resistance-related conditions such as diabetes, metabolic syndrome and NAFLD (Pani et al., 2020; Croze and Soulage, 2013). Likewise, myo-inositol phosphoglycans mediate hormonal signaling transduction, glucose, and lipid metabolism and thereby improve plasma lipid profile and reduce hepatic lipid accumulation (Pani et al., 2020; Tabrizi et al., 2018; Antonowski et al., 2019). Moreover, we previously reported positive effects of 8 weeks myo-inositol supplementation on body weight, lipid profile, glycemic response, blood pressure, and liver function in patients with NAFLD (Arefhosseini et al., 2023; Arefhosseini et al., 2023).

Current evidence suggests the probable effects of INSs on the up-regulation of the SIRT-1 gene, increased levels of sirtuin protein in blood and adipose tissue as well as the subsequent improvement in atherogenesis and oxidative stress as predominant extra-hepatic features of NAFLD (Koh et al., 2018; Sherkhane et al., 2022). While, there is only limited and conflicted knowledge regarding the exact effects of INSs on SIRT-1 downstream genes, especially on the activation or suppression of low-density lipoprotein receptor (LDLR), liver X receptor (LXR), and proprotein convertase subtilisin/kexin type 9 (PCSK9) genes (Shokrpour et al., 2019). Hence, considering the lack of interventional studies regarding this issue, the present randomized clinical trial aimed to assess the effects of myo-inositol supplementation on SIRT-1 pathway gene expression level, atherogenic indices, and hematological parameters in patients with NAFLD.

2. Materials and methods

2.1. Setting of the study and patients

47 obese patients with NAFLD enrolled in the current double-blinded placebo-controlled randomized clinical trial, in order to examine the effects of myo-inositol supplementation on SIRT-1 pathway genes mRNA expression as well as atherogenic and hematological parameters. The study was conducted in accordance to the Declaration of Helsinki. Trials protocol was approved by the Ethics Committee of Research Vice-Chancellor of Tabriz University of Medical Sciences, Tabriz, Iran (Ethics code: IR.TBZMED.REC.1400.1139), and also was registered in the Iranian Registry of Clinical Trials (IRCT20190724044322N2). In addition, the aims of the study were explained and an informed consent form was signed by the participants at baseline. Volunteers were recruited from clinics and polyclinics of Tabriz University of Medical Sciences, Tabriz, Iran. The study was conducted on obese patients with NAFLD, aged 18–55 years with a body mass index of 30–40 Kg/m² and grade I and grade II hepatic steatosis. Ultrasonography (Sonoace X4 Medisio, South Korea) was used for the diagnosis of NAFLD in a fasting state and the degree of liver steatosis was classified into; healthy, grade I, and grade II, assessed by a single radiologist according to Hamaguchi et al. (Hamaguchi et al., 2007) standard criteria. Exclusion criteria of the study were: alcohol drinker or smoker, pregnancy, lactation, menopause, taking any type of drugs/supplements, anti-hyperlipidemic, anti-diabetic, anti-inflammatory, and antibiotic medicines or following specific diets during 3 months before the study and co-morbidity with other disorders e.g. liver disease, infections and metabolic disease including type two diabetes, hypertension, and cancer. Patients who had any changes in their health status or taking medications as well as experiencing any side effects during the trial were withdrawn.

2.2. Sample size

Based on our extensive search, due the lack of human studies conducted on SIRT-1 pathway genes expression and fold-changes, sample size was estimated based on mean (standard deviation (SD)) of serum triglyceride (Livak & Schmittgen, 2001) used from a previous clinical trial on the effects of D-pinitol (Shokrpour et al., 2019) on metabolic and anti-oxidative response of patients with NAFLD considering type I (α) and type II (β) errors of 0.05 and 0.20 (confidence interval = 95 % and power = 80 %) in two-sided tests, using power analysis and sample size software (PASS; NCSS, LLC, US) (Lee, Lim, Kwon, & Kwon, 2019). The calculated sample size was 18, which was increased to 22 after considering a probable drop-out rate of 20 %.

2.3. Study procedures and intervention

A research assistant not otherwise involved in the study without any contact with the participants, randomly allocated the patients with NAFLD into each of the myo-inositol and placebo-receiving arms with a ratio of 1:1, using Random Allocation Software (Ismail et al., 2022) and randomized block procedure of size 3 [gender (female vs male), age (18–35 yrs. vs 36–55 yrs.) and Body mass index (<35 kg/m² vs $\geq$ 35 kg/m²)]. The researcher, assessors, and participants were blind to the assignment. The assignment was concealed from the researcher before randomization for intervention. Both of the study groups were advised to follow healthy dietary recommendations for weight loss and asked to maintain their usual lifestyle. Dietary recommendations were based on a balanced intake of complex carbohydrates, legumes, vegetables, and dairy as well as high-fiber food besides the moderate intake of water. Moreover, the daily intake of simple sugar, oil, and fat was limited (Evert et al., 2014). Indeed, supplement (myo-inositol) and placebo (maltodextrin) sachets (assigned an identical 3-digit code for each of the arms) were prepared and packaged by the research assistant in hygienic conditions. Both sachets contained 2 g of the mentioned ingredients,

distinctly. The myo-inositol and maltodextrin sachets were completely identical in all other aspects, e.g. size and color.

Regarding the intervention, patients in the myo-inositol group (N = 23) received 2 g of myo-inositol powder (Wholesale Health Connection, China) twice a day, and those in the placebo group (N = 24) received 2 g of maltodextrin powder twice a day for eight weeks, respectively. Myo-inositol and maltodextrin powders were consumed by dissolving in a glass of water and drunk 30 min before lunch and dinner. Furthermore, through the biweekly visits, body weight was assessed, sachets were delivered to the patients and they discussed with the researcher any inconvenience with the supplements or dietary recommendations. The compliance rate was assessed by counting the remaining sachets and returning unused sachets. Consuming $\geq 90\%$ of the sachets was considered as compliance.

2.4. Outcome measures

At the beginning of the trial demographic questionnaire was collected for the patients. Then, international physical activity questionnaire form was completed through a face-to-face interview to assess physical activity level (PAL) via recording the time that the respondent spent for each of the determined intensity-varied activities during the past week ([Committee IR. Guidelines for data processing and analysis of the International Physical Activity Questionnaire \(IPAQ\)-short and long forms. <http://www.ipaq.ki.se/scoring.pdf>. 2005, 2005](#)). PAL was reported by the calculated total of Metabolic Equivalent of Task (MET-minutes/week) score according to the manual ([Committee IR. Guidelines for data processing and analysis of the International Physical Activity Questionnaire \(IPAQ\)-short and long forms. <http://www.ipaq.ki.se/scoring.pdf>. 2005, 2005](#)). Indeed, energy intake was also assessed by completing a 3-day food recall (2weekdaysandaweekend) by face-to-face interview. Data on food and beverage intake were analyzed using Nutritionist IV software foods (First Databank, San Bruno, CA, USA) modified for Iranian. The patient's weight and height were measured using a Seca stadiometer (Hamburg, Germany) and a calibrated scale based on the standard protocol to the nearest 100 g and 0.5 cm, respectively. Subsequently, the Body mass index was calculated as weight (Kg) divided by height squared (m^2). Physical activity, energy intake, liver ultrasonography as well as anthropometric assessments were performed at baseline and repeated after 8 weeks.

2.5. Blood sampling and biochemical measurements

A venous blood sample of 10 mL was taken from each patient after 12–14 h of overnight fasting at baseline and end of the trial. Drawn blood samples were separated into two tubes to perform molecular assay (using 5 ml whole blood in EDTA-tubes) forthwith collection and biochemical assay, respectively. The rest of the blood samples were collected in vacuumed gel separator tubes, serum was separated through centrifuge and stored at -80°C until assays.

2.5.1. Atherogenic indices

At the end of the trial, serum total cholesterol (TC), TG and high-density lipoprotein cholesterol (HDL-c) were measured via enzymatic methods using colorimetric technique, by commercial kits (Pars-Azmoon Co., Tehran, Iran). Low-density lipoprotein cholesterol (LDL-c) was calculated by the Friedelwald equation and non-HDL-c was estimated by subtracting HDL-c from TC ([Abid, Abid, & Abid, 2021](#)). Furthermore, atherogenic indices including TC/HDL-c, TG/HDL-c, LDL-c/HDL-c, and non-HDL-c/HDL-c were calculated and the atherogenic index of plasma was estimated based on $\log(\text{TG}/\text{HDL-c})$ formula ([Ismail et al., 2022](#)).

2.5.2. Hematological parameters

At the beginning and after 8 weeks, complete blood count provided information regarding white blood cell count (WBC), red blood cell

count (RBC), hemoglobin (Hgb), hematocrit (HCT), mean corpuscular volume (MCV) and platelet count. Hematological parameters were evaluated by using a Hematology Analyzer (Nihon Kohden Celltac alpha MEK-6510) on the remained whole blood in EDTA tubes. Red cell distribution width-coefficient of variation (RDW-CV) and monocyte-to-HDL-c ratio (MHR), as novel inflammatory indices, were also obtained from complete blood count findings ([Wang et al., 2022](#)).

2.6. Molecular assay

2.6.1. Isolation of peripheral blood mononuclear cells (PBMCs)

As mentioned, in a fasting state 5 ml of fresh venous blood samples taken from patients were collected in vacuum collection EDTA-tubes (Vacutainer K2E). Using density gradient centrifugation, ficoll-Histopaque solution gradient (Ficoll-Paque, GmbH, Germany) was applied to isolate PBMCs from anticoagulant-treated blood. Ficoll-Paque density gradient centrifugation involves layering anticoagulant-treated and diluted blood on a Ficoll-Paque solution and centrifuging it. After centrifugation, erythrocytes and granulocytes settle to the bottom layer.

2.6.2. RNA extraction

Total RNA was extracted from PBMCs and purification was performed using Ambion Trizol LS reagent (Thermo Fisher Scientific, USA), based on the manufacturer's orders. DNaseI (Gene All, Seoul, Korea) was also used for the elimination of DNA contamination and subsequently, the quality and quantity of the extracted RNA was assessed. Gel electrophoresis assay was carried out in order to evaluate RNA integrity. Moreover, using a NanoDrop spectrophotometer (NanoDrop One/OneC, Thermo Scientific), the relative absorbance ratio at A260/280 and A260/230 was determined to confirm the quantity and purity of the extracted RNA.

2.6.3. Complementary DNA synthesis

Complementary DNA was synthesized from total isolated RNA through polymerase chain reaction and by using random hexamer primer, moloney murine leukemia virus, dNTP mix, RNase inhibitor, and complementary DNA synthesis buffer (BioFact™ RTase, South Korea) according to manufacturer's protocol.

2.6.4. RT-qPCR for genes

Quantitative detection of mRNA in samples was done using SYBR Green Master mix (BioFACT™ 2X Real-Time PCR Master Mix, South Korea) for reverse transcription quantitative real-time polymerase chain reaction (RT-qPCR) technique and the expression levels of SIRT-1, endothelial nitric oxide synthase (eNOS), liver X receptor (LXR), LDL receptor (LDLR), proprotein convertase subtilisin/kexin type 9 (PCSK9) and lectin-like oxidized-low-density lipoprotein receptor-1 (LOX-1) mRNA were measured by Bio-Rad IQ5 system (Bio-Rad, USA). RT-qPCR was conducted at 95c for 30 s, followed by 40 cycles at 95c for 5 s, 60c for 30 s, and 72c for 25 s. After the completed RT-qPCR, melting curves were identified at 95c to verify the specificity. Primer sequences (sense and antisense) were designed with the assistance of the Oligo v7.56 software (Bioneer Co. Chungwon, South Korea) and NCBI, summarized in [Table 1](#). β -actin, as a housekeeping gene, was used to normalize the mRNA expression of target genes. Relative expression was quantified based on the changes in cycle threshold and fold change (FC) was estimated using $2^{-\Delta\Delta\text{CT}}$ formula ([Livak & Schmittgen, 2001](#)). The assay was performed in triplicate.

2.7. Study endpoints

In the present study, changes in the mRNA expression levels of SIRT-1, eNOS, LXR, LDLR, PCSK9, LOX-1 in the PBMCs were considered as primary endpoints, while changes in atherogenic, hematological, and inflammatory parameters were defined as the secondary outcomes.

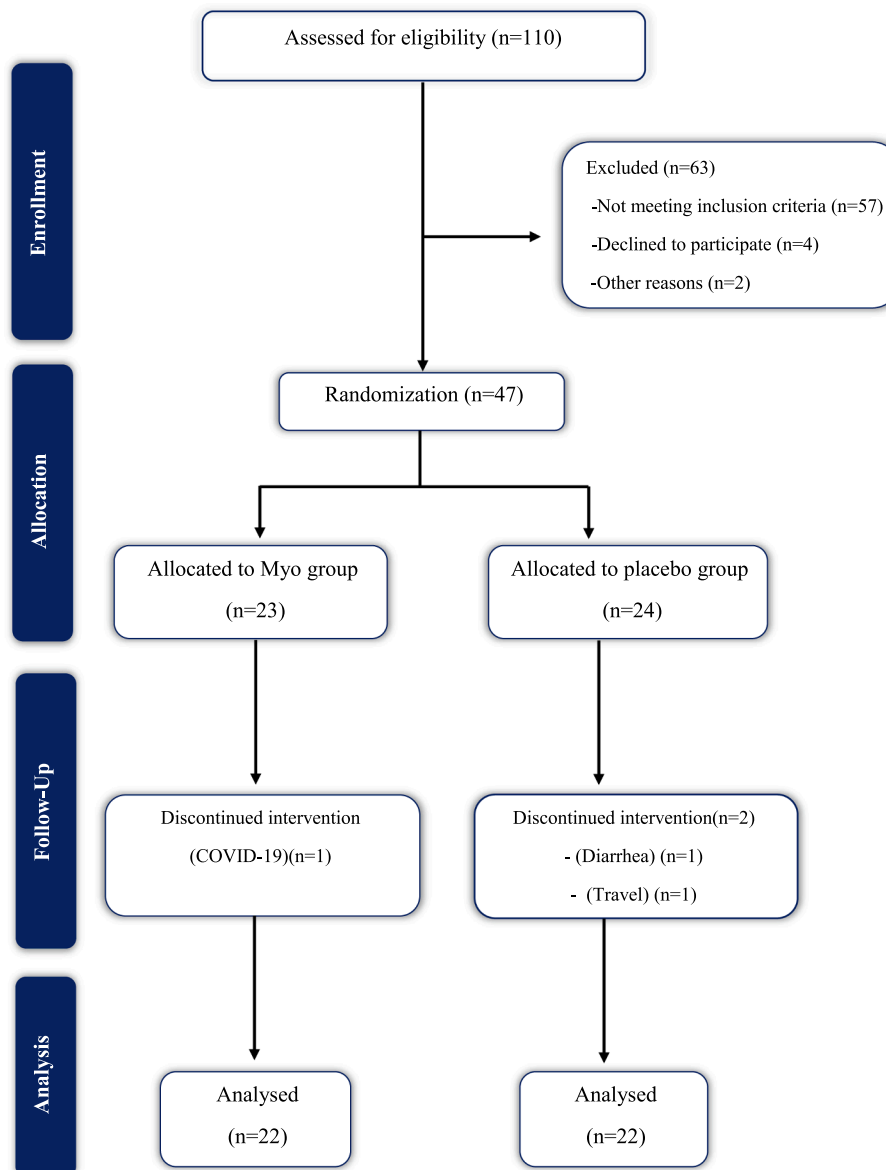
Table 1

Specific primers used for reverse transcription quantitative real-time PCR (RT-qPCR).

β-actin	Forward GGTGAAGGTGACAGCAGT Reverse TGGGGTGGCTTTTAGGAT
SIRT-1	Forward TATACCCAGAACATAGACACG Reverse CAGCTTCACAGTCAACTT
eNOS	Forward GCCCGGACTTCATCAACCAG Reverse CCTCTTGAAGCCGCTGTTCGT
LXR	Forward TTTCACTGAGCTGGCCATCGTC Reverse ATGTCTCCAGAAGCATCACCT
LDLR	Forward GAAATGATCTGGCTGAATCCG Reverse CCAAGGTAACCGGGTGT
PCSK9	Forward CTTGAAGTTGCCCATGTCGA Reverse CTGGTATTTCATCCGCCGGTA
LOX-1	Forward GACTCTACCACTTACGGACA Reverse TAGCTTACAGAAAAGACCACT

2.8. Statistical methods

We applied IBM SPSS Statistics software, version 23 (IBM Corp., Armonk, N.Y., USA) in order to enter and analyze data. Descriptive statistics and the normal distribution of data were assessed using Kolmogorov-Smirnov test. Quantitative and qualitative data were demonstrated as mean (standard deviation (SD)) and frequency (percentage), respectively. An intention-to-treat approach was followed and the difference in the parameters throughout the study was obtained using the following formula: difference = baseline measure – endpoint measure. At baseline, between-group differences were assessed using two independent samples *t*- and Mann-Whitney *U* tests regarding continuous parameters, while the Chi-square test was applied for categorical parameters. Intra-group changes after 8 weeks were evaluated by paired samples *t*-test and Wilcoxon signed-rank test, however, inter-group changes at the end of the trial were tested using independent samples *t*-test and Mann-Whitney *U* test. Analysis of covariance (ANCOVA) was run to compare the two groups at the end of the study by adjusting for the potential confounders (baseline values and energy intake). Mean difference (MD) and 95 % Confidence interval (95 % CI)

**Fig. 1.** Study flow diagram.

were also reported and the statistically significant level was considered at $P < 0.05$.

3. Results

3.1. Demographics

Fig. 1 illustrates the study flow diagram. The current randomized clinical trial involved 47 obese adults with NAFLD. While 44 patients completed the trial (22 male/ 22 female). Three patients, one in the myo-inositol group and two in the placebo group left the trial for reasons unrelated to the intervention. No adverse effects were exhibited following supplementation in the study groups. The general characteristics of the study participants are illustrated in Table 2. The majority of the participants were married and diagnosed with grade II liver steatosis. The mean age of patients in myo-inositol and placebo groups was 35.71 and 38.19, respectively. No significant difference was found between the groups in baseline characteristics, specifically in terms of lipid profile (Table 2).

3.2. Expression of SIRT-1 pathway genes in PBMCs

We examined whether SIRT-1, eNOS, LXR, LDLR, PCSK9, and LOX-1 genes change in expression between the myo-inositol-supplemented group and the placebo group (Fig. 2). The difference in the relative expression of LOX-1 mRNA between the groups was statistically significant ($p = 0.045$) based on the ANCOVA test, after adjusting for the potential confounding variables (baseline values and energy intake). For comparison of the expression level changes of studied genes with unchanged states (steady-state expression), one sample t -test was used. Significant increases in FC of eNOS ($p < 0.05$) were seen in both myo-inositol (FC: 2.93 ± 0.51 ; $p = 0.001$) and placebo group (FC: 2.58 ± 0.47 ; $p = 0.004$). Beyond that, the increment in FC of LDLR was seen in only the myo-inositol group (FC: 2.40 ± 0.43 ; $p = 0.004$). These changes show that the expression of these two genes has increased compared to

Table 2

General characteristics of the participants.

Variable	Myo (N = 22) Mean (SD)	Placebo (N = 22) Mean (SD)	p*
Age (yr.)	35.71 (8.87)	38.19 (6.38)	0.591*
Weight (Kg)	96.29 (15.63)	96.12 (15.67)	0.971*
BMI (Kg/m ²)	34.11 (3.65)	34.12 (3.60)	0.990*
PAL (MET- min/wk)	548.08 (317.29)	462.99 (342.30)	0.378 [†]
TC (mg/dL)	193.43 (41.66)	182.71 (26.15)	0.325*
TG (mg/dL)	183.83 (96.65)	154.67 (58.25)	0.245*
HDL-c (mg/dL)	43.50 (10.59)	44.19 (8.99)	0.823*
LDL-c (mg/dL)	113.16 (32.17)	107.60 (23.61)	0.527*
	%	%	
Gender			
Female	47.6	52.4	0.758**
Male	52.4	47.6	
Marital status			
Single	19	9.5	0.663**
Married	81	90.5	
Educational level			
<University degrees	57.1	33.3	0.121**
>University degrees	42.9	66.7	
Liver steatosis			
Grade I	42.9	42.9	1.000**
Grade II	57.1	57.1	

Myo, Myo- inositol; BMI, Body mass index; PAL, Physical activity level; MET, Metabolic equivalent of task; TC, Total cholesterol; TG, Triglycerides; HDL-c, High-density lipoprotein cholesterol; LDL-c, Low-density lipoprotein cholesterol Mean (SD) is presented for data.

* p value for continuous variables using Independent sample t -test; **p value for categorical variables using Chi square test; [†]p value for non-parametric continuous variables using Mann-Whitney U test.

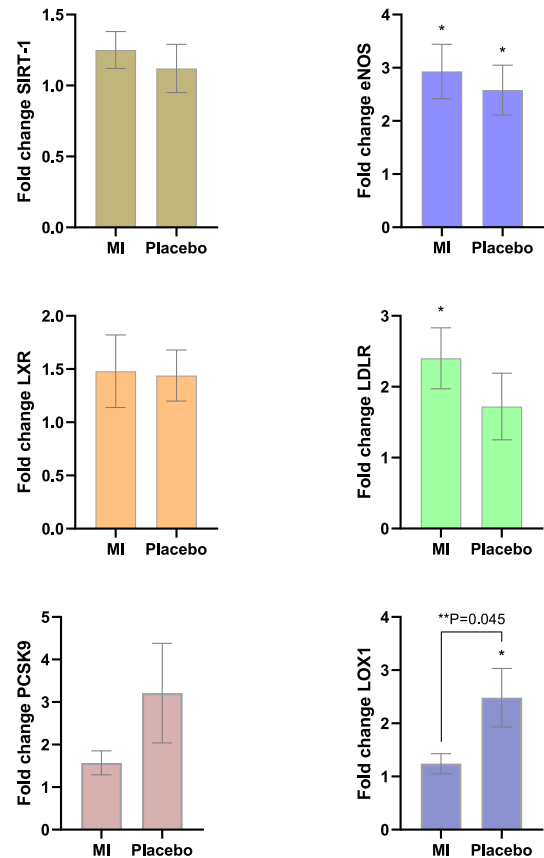


Fig. 2. Result of intervention. Fold change of SIRT-1, eNOS, LXR, LDLR, PCSK9 and LOX-1 mRNAs expression. Values are the mean of fold change $\pm$ s.e.m. Data analysis was done using the One Sample T-test and ANCOVA. * $P < 0.05$ vs. steady-state level. ** $P < 0.05$ vs. Placebo. $P < 0.05$ is statistically significant. Abbreviations: SIRT-1, Sirtuin-1; eNOS, Endothelial nitric oxide synthase; LXR, Liver X receptor; LDLR, Low density lipoprotein receptor; PCSK9, Proprotein convertase subtilisin/kexin type 9; LOX-1, Lectin like oxidized- low-density lipoprotein receptor-1.

the steady-state level in the mentioned groups. Based on the FC formula: $(\text{Efficiency}_{\text{target}})^{\Delta\text{CT}_{\text{target}}(\text{control} - \text{sample})} / (\text{Efficiency}_{\text{reference}})^{\Delta\text{CT}_{\text{ref}}(\text{control} - \text{sample})}$ (Livak & Schmittgen, 2001). FC equal to one is considered a steady-state level. Considering LOX-1 mRNA expression, its expression has decreased insignificantly in the myo-inositol group (FC: 1.24 ± 0.19 ; $p = 0.243$), compared to its significant increase in the placebo group (FC: 2.48 ± 0.55 ; $p = 0.014$) (Fig. 2).

3.3. Atherogenic indices

As shown in Table 3, comparison in atherogenic indices before and after supplementation in both groups revealed significant reductions in cholesterol-related atherogenic indices, including non-HDL-c, TC/HDL-c, LDL-c/HDL-c and Non-HDL-c/HDL-c in myo-inositol group compared with baseline values. Nevertheless, inter-group changes did not reach to statistically significant levels.

3.4. Hematological and inflammatory parameters

Changes in hematological and related inflammatory parameters of the study participants after 8 weeks of intervention is presented in Table 4. At the end of the study changes in WBC, Hgb, HCT, and MCV were in a decreasing trend in myo-inositol group. Compared to baseline values, WBC and Hgb significantly reduced in myo-inositol group, while, HCT levels markedly increased in the placebo group. Post-intervention, expecting HCT levels ($p = 0.043$), the inter-group differences in other

Table 3
Atherogenic indices at baseline and 8 weeks after the intervention.

Variable	Myo (N = 22) Mean (SD)	Placebo (N = 22) Mean (SD)	p
Non- HDL-c (mg/dl)			
Baseline	149.92 (35.92)	138.53 (23.40)	0.330**
End	133.46 (34.49)	132.43 (29.52)	0.264***
MD (95 % CI)	16.46 (3.89, 29.03)	6.10 (−1.51, 13.71)	
p*	0.013	0.11	
TC/HDL-c			
Baseline	4.54 (0.86)	4.25 (0.79)	0.254**
End	4.02 (0.93)	4.12 (0.89)	0.332***
MD (95 % CI)	0.52 (0.13, 0.91)	0.12 (−0.29, 0.54)	
p*	0.011	0.538	
TG/HDL-c			
Baseline	4.29 (2.29)	3.56 (1.26)	0.207**
End	3.79 (2.07)	3.29 (1.32)	0.831***
MD (95 % CI)	0.50 (−0.55, 1.56)	0.26 (−0.05, 0.59)	
p*	0.332	0.099	
LDL-c/HDL-c			
Baseline	2.69 (0.84)	2.54 (0.75)	0.549**
End	2.26 (0.74)	2.47 (0.86)	0.411***
MD (95 % CI)	0.42 (0.10, 0.74)	0.07 (−0.33, 0.48)	
p*	0.013	0.715	
Non- HDL-c/HDL-c			
Baseline	3.54 (0.86)	3.25 (0.79)	0.254**
End	3.02 (0.93)	3.12 (0.89)	0.332***
MD (95 % CI)	0.52 (0.13, 0.91)	0.12 (−0.29, 0.54)	
p*	0.011	0.538	
AIP			
Baseline	0.57 (0.23)	0.52 (0.16)	0.413**
End	0.52 (0.23)	0.48 (0.18)	0.802***
MD (95 % CI)	0.05 (−0.05, 0.16)	0.04 (−0.02, 0.10)	
p*	0.292	0.16	

Myo, Myo- inositol; Non-HDL-c, Non- high density lipoprotein cholesterol; TC/HDL-c, Total cholesterol to high-density lipoprotein cholesterol ratio; TG/HDL-c, Triglycerides to high-density lipoprotein cholesterol ratio; LDL-c/HDL-c, Low-density lipoprotein cholesterol to high-density lipoprotein cholesterol ratio; Non-HDL-c/HDL-c; Non- high density lipoprotein cholesterol to high density lipoprotein cholesterol ratio; AIP, Atherogenic index of plasma Mean (SD) and Mean Difference (95 % CI) are presented for data.

* p value for paired- *t* test; ** p value for Independent samples *t*-test; *** p value for ANCOVA test (adjusted for baseline values and energy intake. Bold values indicates statistically significant ($p < 0.05$).

hematological parameters remained insignificant after adjusting for potential confounders. Regarding CBC-derived inflammatory parameters, myo-inositol supplementation significantly decreased MHR, after 8 weeks compared to baseline as well as in comparison with the changes in the placebo group ($p = 0.043$ and $p = 0.031$, respectively), while no significant differences was documented between the two groups, in terms of RDW-CV (Table 4).

4. Discussion

For the first time, the current randomized, double-blind randomized clinical trial investigated the effects of 8 weeks myo-inositol supplementation plus dietary recommendations on SIRT-1 pathway genes expression, atherogenic indices, and hematological parameters in obese patients with NAFLD. Our results mainly encompassed the impacts of myo-inositol on the slight regulation of SIRT-1 downstream genes, followed by marginal improvements in atherogenic and hematological parameters.

As mentioned, SIRT-1 cascade is involved in the inter-connection of NAFLD, dyslipidemia, and atherogenesis (Ding, Bao, & Deng, 2017). Consequent to the activation of SIRT-1 cascade, endothelial function, and OS are improved through SIRT-1 and eNOS stimulation (Feng et al., 2018). On the other side, the up-regulation of LDLR and LXR, besides PCSK9 suppression results in the improvement of dyslipidemia and LOX-1 down-regulation leads to the prevention of foam cell formation (Feng

Table 4
Hematologic and inflammatory parameters at baseline and 8 weeks after the intervention.

Variable	Myo (N = 22) Mean (SD)	Placebo (N = 22) Mean (SD)	p
WBC ($\times 10^3$ /mm ³)			
Baseline	7.7 (1.6)	7.4 (1.7)	0.479**
End	6.8 (1.3)	7.0 (1.6)	0.293***
MD (95 % CI)	0.92 (0.45, 1.39)	0.4 (−0.2, 0.9)	
p*	0.001	0.183	
RBC ($\times 10^6$ /mm ³)			
Baseline	5.1 (0.6)	5.0 (0.5)	0.570**
End	5.1 (0.5)	5.1 (0.4)	0.358***
MD (95 % CI)	0.05 (−0.11, 0.20)	−0.06 (−0.16, 0.04)	
p*	0.53	0.249	
Hgb (g/dL)			
Baseline	14.7 (1.9)	14.5 (1.5)	0.711**
End	14.2 (1.4)	14.5 (1.4)	0.109***
MD (95 % CI)	0.48 (0.02, 0.94)	0.03 (−0.28, 0.34)	
p*	0.042	0.826	
HCT (%)			
Baseline	43.3 (4.9)	42.7 (3.8)	0.595**
End	42.7 (4.0)	43.4 (3.4)	0.043***
MD (95 % CI)	0.67 (−0.46, 1.80)	−0.72 (−1.38, −0.07)	
p*	0.23	0.032	
MCV (fL)			
Baseline	84.8 (6.2)	85.1 (4.2)	0.836**
End	84.1 (6.3)	85.5 (4.5)	0.185***
MD (95 % CI)	0.65 (−0.67, 1.98)	−0.36 (−1.47, 0.75)	
p*	0.318	0.506	
PLT ($\times 10^3$ /mm ³)			
Baseline	253.1 (49.8)	242.0 (41.52)	0.408**
End	245.8 (57.0)	245.6 (48.8)	0.535***
MD (95 % CI)	7.3 (−12.8, 27.4)	−3.6 (−15.9, 8.7)	
p*	0.461	0.553	
RDW- CV (%)			
Baseline	13.27 (0.89)	13.53 (0.89)	0.305**
End	13.58 (1.23)	13.18 (0.88)	0.197***
MD (95 % CI)	−0.31 (−0.81, 0.19)	0.36 (−0.11, 0.82)	
p*	0.206	0.124	
MHR			
Baseline	12.6 (4.6)	10.3 (5.5)	0.043^{††}
End	9.0 (3.4)	9.6 (3.4)	0.031***
MD (95 % CI)	3.6 (2.3, 5.0)	0.7 (−1.4, 2.8)	
p [†]	<0.001	0.932	

Myo, Myo-inositol; WBC, White blood cell count; RBC, Red blood cell count; Hgb, hemoglobin; HCT, Hematocrit; MCV, Mean cell volume; PLT, Platelet count; RDW-CV, Red blood cell distribution width; MHR, Monocyte to-HDL ratio. Mean (SD) and Mean Difference (95 % CI) are presented for data.

*p value for paired- *t* test; **p value for Independent samples *t*-test; ***p value for ANCOVA test (adjusted for baseline values and energy intake); [†] P value for Related samples Wilcoxon signed rank test; ^{††} P value for Mann- Whitney *U* test. Bold values indicate statistically significant ($p < 0.05$).

et al., 2018). In all, the mentioned changes emerge as the alleviation of metabolic and atherogenic aspects of NAFLD (Zhang et al., 2023; Feng et al., 2018). In our study, myo-inositol supplementation failed to increase SIRT-1 mRNA expression, while eNOS was significantly up-regulated almost two- times more than the steady-state levels in both groups. Regarding SIRT-1, our findings are in contrast with the experimental studies that reported the effectiveness of INs (myo-inositol and DP) on SIRT-1 up-regulation in renal disorders (Koh et al., 2018; Sherkhane et al., 2022). In addition, Lopez-Domenech et al. (López-Domènech, 2018) reported enhanced SIRT-1 protein in PBMCs and adipose tissue of subjects with obesity who underwent 12-week treatment with DP-enriched beverage (containing 2.29 g of INs) (López-Domènech, 2018). On the other side, in terms of eNOS expression, our results are partially in line with previous studies that reported stimulated eNOS expression after DP exposure in cellular and animal models (Siracusa et al., 2022; Moreira et al., 2018). More precisely, DP administration in mice resulted in the promoted phosphorylation levels of eNOS activation sites, via calcium-dependent mechanisms which led

to hypotensive effects on the mesenteric artery (Moreira, Silva, Silva, Lemos, & Cortes, 2018). It is suggested that eNOS could also be phosphorylated through the stimulation of the phosphoinositol-3-kinase/protein kinase-B pathway after INS administration (Siracusa et al., 2022; D'Orta et al., 2017). Activated eNOS, exerts anti-oxidant features by increasing NO bioavailability, reducing reactive oxygen species, endothelial apoptosis and preventing atherogenesis (Koh et al., 2018; Zhang et al., 2017).

Of note, LDLR, PCSK9, and LXR as downstream genes of the SIRT-1 pathway are considered great targets in the management of dyslipidemia (Shokrpour et al., 2019). The inhibition of PCSK9 and activation of LDLR predominantly regulate blood TC homeostasis (Shokrpour et al., 2019). LXR, as a sterol sensor, also stimulates hepatic TC efflux resulting in decreased serum TC, TG as well as atherosclerotic plaques (Feng et al., 2018). In the present study, 8 weeks of supplementation with 4 g/day of myo-inositol, markedly increased LDLR expression levels compared with the basal state only in myo-inositol group, while did not affect PCSK9 and LXR gene expression. Despite the lack of studies regarding PCSK9 and LDLR expression, our findings are opposed to Shokrpour et al. (Shokrpour et al., 2019) which represented unchanged FC of LDLR gene in PBMCs of women with polycystic ovary syndrome who consumed 4 g/day myo-inositol + 400 mg/day folic acid compared with 1500 mg/day metformin, for 12 weeks (Shokrpour et al., 2019). Our findings could also be positively attributed to the trial conducted by Lee et al. (Lee et al., 2019) who supplemented 90 patients with NAFLD by 300 mg/day and 500 mg/day of DP for 12 weeks. The observed small reduction in serum TC and LDL-c in the DP group as well as non-significant differences between DP groups and placebo was similar to our results (Lee et al., 2019). LDLR up-regulation predominantly regulates lipidemia by fine-tuning lipoprotein removal from the circulation, resulting in reduced LDL-c (Shokrpour et al., 2019). Improved TC components emerge as favorable atherogenic status in NAFLD.

The current body of evidence enlightens the interplay between oxidized- LDL and endothelial dysfunction in the pathogenesis of atherogenesis (Feng et al., 2018). LOX-1, an oxidized- LDL scavenging receptor, exacerbates inflammatory and oxidative pathways as well as adhesion molecule production leading to atherosclerosis (Chen, Tan, Yu, Lu, & Deng, 2022). The main finding of our trial was the significant inhibition of LOX-1 mRNA expression in myo-inositol group, compared with the increased levels in the placebo group. In line with our findings, a cell line study documented the favorable effect of DP on decreasing the expression of LOX-1 in a model of oxidized-LDL-induced endothelial dysfunction (Chen et al., 2022). Following them, An et al. (An, Li, Liu, Fan, & Tian, 2020) in an animal study on spinal cord injury, illustrated the effects of DP (in 10 mg/Kg/day to 30 mg/Kg/day doses) on improving OS, inflammation, and a significant dose-dependent decrease in LOX-1 levels (An et al., 2020). The underlying mechanism could be related to the anti-inflammatory properties of INSs which result in down-regulated nuclear factor-kappa B and tumor necrosis factor- α which reduce LOX-1 expression and oxidized-LDL uptake (Arefhosseini et al., 2023; Stein et al., 2010). Moreover, SIRT-1 activation deacetylates RelA/P65, inhibits nuclear factor kappa-B, and thereby down-regulates LOX-1 and KLF2, as an athero-protective factor, and prevents foam cell formation (Stein et al., 2010).

As mentioned, our team previously reported the anti-hyperlipidemic effects of myo-inositol supplementation in patients with NAFLD by lowering serum TG, TC, LDL-c and increasing HDL-c levels (Arefhosseini et al., 2023). It is noteworthy that atherogenic indices are considered as a the mirror of the metabolic interactions of lipid fractions and exhibit higher prediction values rather than lipid profiles in cardiovascular disease (Millán et al., 2009). Adding strength to the previous findings, we documented significantly decreased TC-related atherogenic indices in the myo-inositol group as well as no marked differences between the two groups, post-intervention. It is suggested that the marginally decreased Non-HDL-c and Non-HDL-c/ HDL-c could reflect the improvements in the major serum lipoproteins as well as decreased

independent risk of NAFLD and cardiovascular disease after 8 weeks of myo-inositol supplementation (Arefhosseini et al., 2023). Our findings were consistent with Kim et al. (Kim et al., 2005) who reported lowered LDL-c/HDL-c levels and cardiovascular risk after 13 weeks of DP (600 mg/day) supplementation in patients with type 2 diabetes. In line with us, they failed to exhibit significant reductions in the TG/HDL-c index (Kim et al., 2005). Despite the limitation of studies on atherogenic indices, a high number of trials have evaluated lipid-lowering features of INSs (Tabrizi et al., 2018). A meta-analysis study on 14 clinical trials on the effects of INSs (mainly myo-inositol and DCI) on metabolic disease illustrated significant reductions in TG, TC, and LDL-c, without affecting serum HDL-c levels (Tabrizi et al., 2018). It should be noted that the anti-hyperlipidemic effects of myo-inositol are the mechanism of its action on atherosclerosis. Myo-inositol in forms INS- containing phosphor glycans, suppresses adenylate cyclase activity and thereby lowers free fatty acid release from the adipocytes (Antonowski et al., 2019; Dinicola et al., 2021). Moreover, INS compounds are involved in enhancing energy expenditure, lipid oxidation, and improving lipid profile through a number of genes including AMP-activated protein kinase, uncoupling protein and peroxisome proliferator-activated receptors (Chatree, Thongmaen, Tantivejkul, Sitticharoon, & Vucenik, 2020). It is hypothesised that the slight activation of LDLR and inhibition of LOX-1 in our study could be responsible for the improvements in atherogenic indices. Fig. 3 illustrates the proposed mechanistic effects of myo-inositol on the progression of atherogenesis and NAFLD.

The current body of evidence suggests hematological changes in patients with NAFLD including higher WBC, Hgb, HCT and MCV levels as well as altered platelet activity (Tutunchi et al., 2020; Öztürk et al., 2018; Seo and Lee, 2022). In addition, the potent role of OS, as a pivotal pathological consequence in NAFLD, on damaging erythrocyte membrane and thereby affecting RBC and RDW levels which contribute to the progression of cardiovascular disease is evident (Tutunchi et al., 2020; Seo and Lee, 2022). Given the intricate association between complete blood count components and NAFLD, we found partial improvements in WBC, Hgb, HCT and MCV as well as significant inhibition of increased HCT levels, only in the myo-inositol group after 8 weeks. Despite the lack of human studies, our results are similar to an animal study on rats with type 2 diabetes, treated with combined inositol hexakisphosphate and inositol in comparison to glibenclamide for 4 weeks (Foster, Dilworth, Sparks, Alexander-Lindo, & Omoruyi, 2019). Excepting RDW levels, they did not find any significant changes in complete blood count findings between the two groups (Foster et al., 2019). We hypothesised that slightly improved Hgb levels after myo-inositol intervention might be related to the increased expression levels of eNOS in our study. eNOS activation leads to nitric oxide production (NO) and enhanced blood flow thereby affecting Hgb and MCV levels (Schechter & Gladwin, 2003). Lowered Hgb levels eliminate the risk of NAFLD as well as arterial stiffness and atherosclerosis (Öztürk et al. (3):71.). Moreover, despite unchanged levels of RDW, we found marked reductions in MHR in the myo-inositol group at the end of the trial. MHR, as a novel indicator, reflects the state of oxidative stress and inflammation and increased MHR levels correlate with a higher risk of cardiovascular disease and fatty liver disease (Wang et al., 2023). The anti-inflammatory effect of myo-inositol could be related to the suppression of tumor necrosis factor- α and its effect on downstream pro-inflammatory cytokines (Arefhosseini, Roshanravan, Asghari, Tutunchi, & Ebrahimi-Mameghani, 2023). Collectively, the slight changes in hematological parameters might be powered by a longer intervention period or assessing higher doses of myo-inositol.

5. Conclusion

Our results mainly encompassed the impact of 4 g/day myo-inositol supplementation for 8 weeks on the SIRT-1 pathway, and atherogenic and hematological parameters in obese patients with NAFLD. A slight up-regulation of eNOS and LDLR mRNA was observed which was

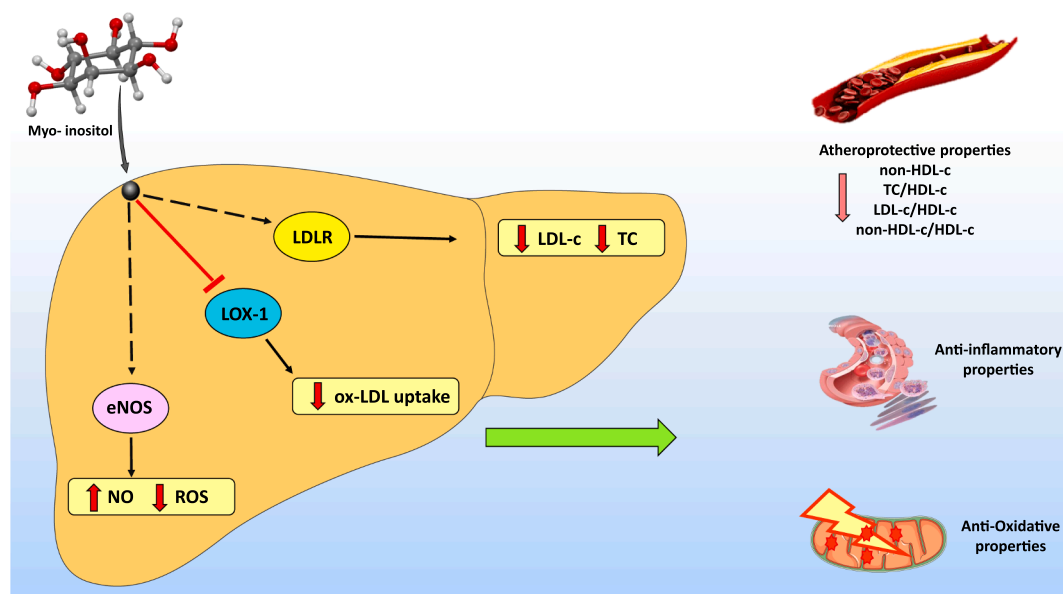


Fig. 3. Proposed mechanistic effects of Myo in the treatment of NAFLD. Myo as a cyclohexanehexol, marginally up-regulates LDLR and eNOS genes and thereby improves lipid profile components, increases NO bioavailability, and reduces ROS production. On the other side, Myo intervention inhibits expression levels of LOX-1 mRNA and ox-LDL uptake. The mentioned molecular changes emerge as slight atheroprotective, anti-inflammatory, patients anti-oxidative changes in obese patients with NAFLD.

consistence with the improvements in lipid homeostasis as well as marginal improvements in TC-related atherogenic indices, especially in terms of Non-HDL-c and Non-HDL-c/ HDL-c. Moreover, myo-inositol supplementation resulted in suppressed expression of LOX-1 mRNAs in PBMCs. Partial improvements in hematological parameters were also documented in obese patients with NAFLD. While further well-designed clinical trials with longer intervention periods, different myo-inositol doses and larger sample sizes are needed.

6. Strengths and limitations

The main strength of the present trial was the assessment of newly diagnosed obese patients with NAFLD, not receiving any treatment as well as evaluating SIRT-1 pathway, atherogenic and hematological parameters after myo-inositol supplementation for the first time on patients with NAFLD. Using ANCOVA test adjusted for potential confounding variables for statistical analysis was another strength of this study. However, several limitations could be considered for this trial. First, serum urinary levels of myo-inositol were not assessed due to funding constraints. The lack of liver biopsy for NAFLD assessment due to ethical considerations and subjective assessment of energy intake were other limitations of the present study.

Ethics approval and consent to participate

All procedures performed in this study were in accordance with the ethical standards of the Ethics Committee of Tabriz University of Medical Science. In this study all subjects signed a consent form and the study protocol was approved by the ethical committee of Tabriz University of Medical Sciences (Ethics code: IR.TBZMED.REC.1400.1139, grant number: 68779).

Authors' contributions

The authors' responsibilities were as follows: SAH, SA and AB helped in data collection and laboratory assessments; SAH wrote the original paper; NR did statistical analysis; NR and MEM contributed to the conception of the article as well as to the final revision of the

manuscript. All authors read and approved the final version of the manuscript.

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CRediT authorship contribution statement

Sara Arefhosseini: Writing – original draft, Investigation. **Mehrangiz Ebrahimi-Mameghani:** Methodology, Formal analysis, Conceptualization. **Samira Asghari:** Methodology, Data curation. **Abolfazl Barzegari:** Validation, Investigation. **Neda Roshanravan:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jff.2024.106197>.

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